Evaluating TimesFM for Multi-Scale Wind Speed Forecasting: A Benchmark Study Against XGBoost

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Abstract. This study investigates the application of TimesFM, a decoder-only foundation model for time series forecasting, to predict wind speed across multiple temporal resolutions using proprietary data from MetService New Zealand. TimesFM's performance is evaluated against the widely used XGBoost model, focusing on its zero-shot forecasting capabilities. The research compares optimal configurations of TimesFM for meteorological datasets and assesses its suitability for specialized forecasting tasks. The results highlight both the potential and current limitations of TimesFM in domain-specific applications such as meteorology.

Keywords: Time-Series Forecasting · Foundational Models · Meteorology · TimesFM · XGBoost.

1 Introduction

Time series data refers to sequences of data points collected at regular time intervals and is widely used in tasks such as content-based querying, anomaly detection, forecasting, clustering, classification, and segmentation [7]. Recent advancements in machine learning and deep learning, especially in natural language processing (NLP) and large language models (LLMs) have also influenced time series modeling [19]. Architectures such as recurrent neural networks (RNNs) [17], convolutional neural networks (CNNs) [5], and transformers [16] have been adopted for time-series forecasting, yet they face persistent challenges.

These challenges include: (1) *data sparseness*, where time series are collected at varying frequencies often lead to gaps that degrade deep model performance; (2) *multimodal integration*, as combining data from multiple sources and frequencies introduces complexity in representation and learning; and (3) *knowledge transfer limitations*, since distinct characteristics such as trend, seasonality, and randomness in different datasets hinder generalisation across domains [18].

Foundational models are designed to handle these issues by generalising across domains particularly newer datasets by pre training and applying knowledge gained from one domain to solve new tasks [9]. Critical applications of weather forecasting in fields such as disaster management, agriculture and energy optimisation has created a focus on accurate weather forecasting.

Traditional machine learning methods like XGBoost have been widely used in meteorological forecasting due to reasonably accurate forecasts with reasonable deployment time. Due to this XGBoost proves to be a great baseline to compare [3]. On the other hand, TimesFM seems to be a great choice to benchmark on weather data due to its less computational demands and promising metrics in their tests[6]. This paper investigates the comparative performance of XGBoost and TimesFM in forecasting meteorological variables using high-dimensional, multivariate data from weather stations.

2 Background

Machine learning (ML) models, especially ensemble techniques like XGBoost, have shown strong performance in capturing non-linear relationships in time-series data. For example, in [3] compared statistical and ML models across various meteorological variables (e.g., temperature, wind, and precipitation) and found that XGBoost achieved high predictive accuracy, with the exception of local anomalies in regions like Swindon.

Deep learning (DL) methods, particularly LSTM-based architectures, show great promise in capturing long-term dependencies in sequential data. Variants such as Stacked-LSTMs and Bidirectional LSTMs have demonstrated significant improvements over simpler models in weather-related tasks, including rainfall prediction [3].

Recent advancements in foundational models introduce a new paradigm. These models are pre-trained on diverse datasets and support zero-shot forecasting, reducing the need for retraining in new domains. TimesFM is a decoder-only transformer meaning given a sequence of input time series divided into patches, the model is optimized to predict the next patch as a function of all past patches [6]. Notably, the model supports smaller input patches and larger output patches, enabling efficient long-horizon predictions while stacked transformer architecture enables multivariate forecasting, as shown in benchmark evaluations across the Monash Archive, Darts datasets, and ETT tasks [6, 11, 20].

In another study, Meyer et al. (2024) presents a benchmark of foundation models like TimesFM, Chronos, and LagLlama against trained from scratch Transformer models on short term household electricity forecasting using data from over 300 households. They found that TimesFM outperformed TFS Transformers in scenarios with longer input sizes, demonstrating its strength in capturing temporal dependencies over extended contexts. However, other foundation models like LagLlama struggled with limited input sizes due to their reliance on lag-based architectures. While Meyer's results underscore the potential of foundation models for time-series forecasting, they also highlight limitations such as sensitivity to input sizes and the risk of information leakage from pre-training datasets [15].

This study builds upon those findings by applying TimesFM to real-world meteorological data from MetService New Zealand and benchmarking it against XGBoost. The goal is to assess whether foundation models offer measurable advantages in operational settings that demand fast, scalable, and accurate forecasting across multiple frequencies.

3 Data Preprocessing

Effective time-series forecasting relies heavily on data preprocessing. This study involved merging datasets of varying frequencies, cleaning and interpolating missing values, and engineering features suitable for machine learning models.

3.1 Data Cleaning and Combining

NWP data in NetCDF format were loaded using the *xarray* library [12, 14]. Observational and forecast datasets were merged on a unified *valid_time* index. Missing values and duplicate timestamps were handled using listwise deletion and retention of the latest WRF model runs. Gaps under four hours were linearly interpolated [4, 1], enabling alignment across hourly (WRF), three-hourly (ECMWF), and high-resolution observed data [10].

3.2 Feature Engineering

Key steps included:

- **Selection:** Top 40 by average importance score across 5 runs were selected using Random Forest importance, ALE plots, and PDPs [2, 8, 13].
- **Transformation:** Temperature values were converted to Kelvin; topographic attributes (latitude, longitude, elevation) were added.
- **Augmentation:** Hour-of-day and month indicators were included to capture temporal seasonality.

3.3 Target Feature Definition

3.4 Target Variable

Wind speed was selected as the prediction target due to its meteorological relevance. Hourly, 10-minute, and derived 3-hour averages were used across datasets, with interpolation applied as needed for alignment.

3.5 Final Dataset Summary

After preprocessing, datasets were shaped as follows:

- 10-minute: (797,135, 45)
- 1-hour: (130,798, 45)
- 3-hour: (43,595, 45)

Training used a 60/20/20 split, while TimesFM used only the last 20% as test data due to its zero-shot nature.

4 Experiments and Results

4.1 Model Overview and Setup

Two models were evaluated: a tree-based baseline model (XGBoost) and a transformer-based foundational model (TimesFM). XGBoost was implemented using the Darts library with default settings, while TimesFM leveraged Google's pre-trained decoder-only architecture for zero-shot time-series forecasting.

Datasets at three temporal granularities—10-minute, hourly, and 3-hourly—were used. For each frequency, models were tested under consistent input-output configurations:

- **XGBoost:** Context length = 30; horizon varied per frequency.
- **TimesFM:** Context length = 512; batch size = 64.

For TimesFM, two hybrid configurations were tested:

1. *xreg + TimesFM:* Regression model applied first, residuals modeled by TimesFM. *xreg + TimesFM* proved to be better at forecasting and thus was the configuration used to benchmark against XGBoost.
2. *TimesFM + xreg:* TimesFM applied first, residuals modeled by regression.

4.2 Evaluation Metrics

Model performance was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the Coefficient of Determination (R^2), computed on test sets with a focus on Station 93110.

4.3 Results

Next we evaluated TimesFM and XGBoost across different frequencies and forecast horizons, focusing on Station 93110 located in Auckland. TimesFM consistently outperformed XGBoost in both accuracy and inference time, with differences becoming more pronounced at higher data granularities (Table 1). At 10-minute frequency, TimesFM achieved a significantly lower MAE (1.74 vs. 2.63) and was over 8x faster due to its zero-shot nature[1]. Additionally, although the target feature was identical across datasets, the hourly dataset generally performed better than three hourly dataset likely due to more data being available.

Effect of Forecast Horizon To assess the models at different frequencies to compare their advantages against the length of forecast from 1 hour till 36 hours in the future. The models were compared with only hourly and three hourly frequency due to high computational demands in forecasting using XGBoost on 10 minute temporal frequency. Figure 1 and 2 show line graph and violin plots respectively visualising a comparison of the values of MAE against the forecast horizon.

[1] Timing for XGBoost includes training and inference; TimesFM uses inference only.

Model	Frequency	Horizon	MAE	MSE	R^2	Time (s)
TimesFM	10-min	3 hrs	1.74	5.35	0.75	96.76
XGBoost	10-min	3 hrs	2.63	13.14	0.74	786.48
TimesFM	Hourly	18 hrs	1.96	7.01	0.70	11.82
XGBoost	Hourly	18 hrs	2.88	13.96	0.71	240.70
TimesFM	3-Hourly	18 hrs	1.94	6.32	0.69	13.05
XGBoost	3-Hourly	18 hrs	3.72	23.34	0.58	46.41

Table 1. Performance comparison of TimesFM and XGBoost across different frequencies at Station 93110.

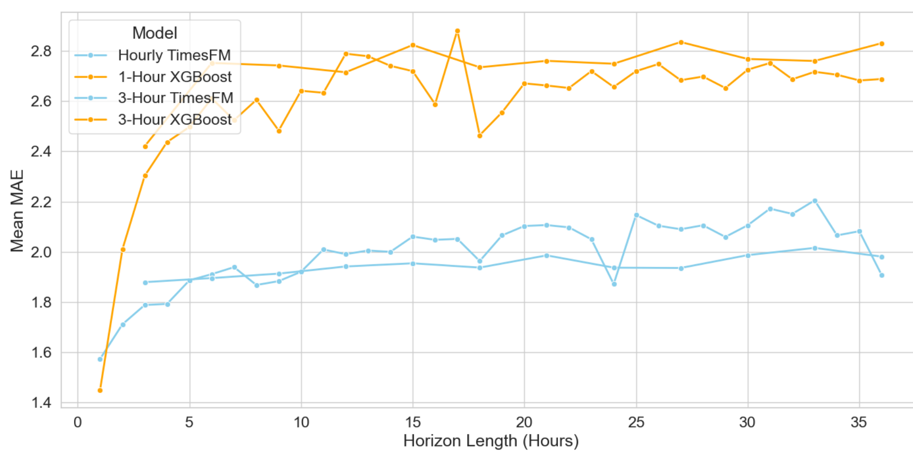

Fig. 1. MAE vs. Horizon Length (Hourly vs Three Hourly TimesFM and XGBoost).

From the line plot in Figure 1, we observe that TimesFM consistently outperforms XGBoost across both hourly and three-hourly frequencies, with an average MAE difference of more than 0.5. An exception is observed at the 1-hour forecast horizon, where XGBoost slightly outperforms TimesFM. Beyond this horizon, TimesFM demonstrates better stability and lower errors.

The violin plot in Figure 2 further supports this trend, showing that TimesFM predictions at three-hourly intervals exhibit tighter distributions. The median MAE is closer to the interquartile range, with shorter tails compared to XGBoost. This suggests reduced variability and more consistent performance from TimesFM. In contrast, XGBoost displays a wider spread in MAE values, indicating increased performance degradation as the forecast horizon extends. This spread highlights the greater sensitivity of XGBoost to longer-term forecasting, while TimesFM appears more robust in handling extended horizons.

Additionally, Figure 1 shows that model performance tends to plateau beyond a 6-hour forecast horizon, likely due to the diminishing influence of recent input features and increasing uncertainty in weather patterns over time.

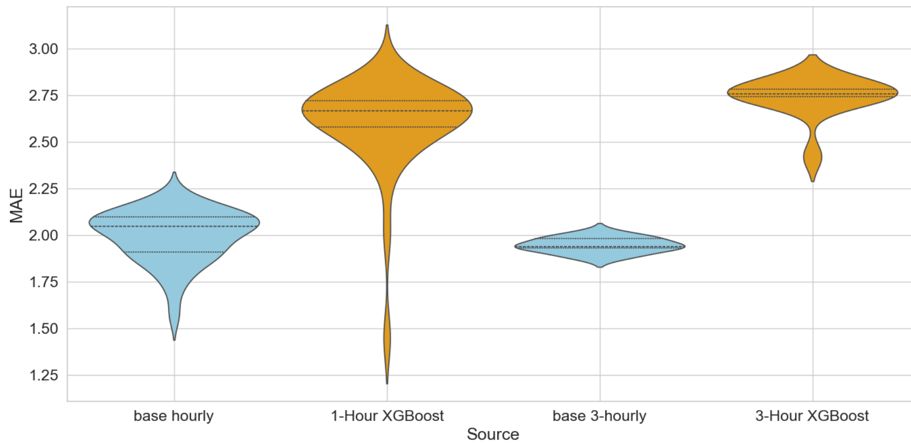

Fig. 2. MAE Distribution across Frequencies (Station 93110).

Hybrid Configuration Impact TimesFM's performance varied significantly based on the hybrid configuration mode. In our experiments, the base TimesFM model outperformed both XGBoost and the *TimesFM + xreg* configuration across all forecast horizons (see Figure 3). The *TimesFM + xreg* mode, which applies a regression model to the residuals of TimesFM predictions, resulted in higher error, especially at longer horizons. This performance drop may stem from residuals produced by TimesFM on our weather dataset likely being irregular or noisy, reducing the effectiveness of downstream regression. These results suggest that the residual structure in our dataset does not favor regression-based refinement, making the base TimesFM configuration more robust and accurate.

XGBoost showed limited benefit from increasing context length; longer historical inputs led to higher training costs without consistent improvements in accuracy.

4.4 General Discussion

TimesFM predominantly outperformed XGBoost, offering both higher accuracy and significantly faster inference, especially at higher data granularities such as the 10-minute frequency. This highlights its suitability for high-resolution forecasting scenarios where retraining is infeasible.

Across forecast horizons, TimesFM remained stable while XGBoost's performance degraded, indicating that TimesFM better captures long-term dependencies. Hybrid configurations further revealed that the base (*xreg + TimesFM*) TimesFM model was more reliable than the *TimesFM + xreg* setup, which likely suffered from noisy residuals that limited regression-based refinement.

Overall, these results underscore TimesFM's robustness for granular, long-horizon forecasting and affirm the importance of configuration choice when applying foundational models.

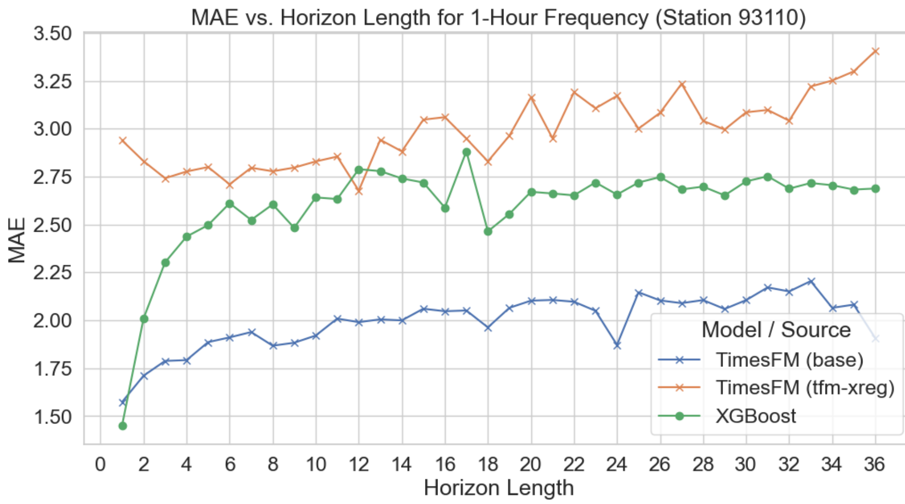

Fig. 3. MAE vs. Horizon Length on hourly frequency (TimesFM(xreg+tfm/base), TimesFM(tfm+xreg) and XGBoost)

5 Conclusion

This study benchmarked TimesFM against XGBoost for wind speed forecasting using meteorological data. TimesFM consistently outperformed XGBoost in both accuracy and inference time, especially for high-resolution and long-horizon tasks. Its zero-shot capability and scalability make it well-suited for operational forecasting.

While TimesFM requires careful configuration and lacks domain-specific fine-tuning, its transformer-based design offers a promising alternative to traditional models in meteorology.

References

1. Anjomshoaa, A., Salmanzadeh, M.: Filling missing meteorological data in heating and cooling seasons separately. International journal of climatology **39**(2), 701–710 (2019)
2. Apley, D.W., Zhu, J.: Visualizing the effects of predictor variables in black box supervised learning models (2019), https://arxiv.org/abs/1612.08468
3. Barrera-Animas, A.Y., Oyedele, L.O., Bilal, M., Akinosho, T.D., Delgado, J.M.D., Akanbi, L.A.: Rainfall prediction: A comparative analysis of modern machine learning algorithms for time-series forecasting. Machine Learning with Applications **7**, 100204 (2022). https://doi.org/https://doi.org/10.1016/j.mlwa.2021.100204, https://www.sciencedirect.com/science/article/pii/S266682702100102X
4. Briggs, A., Clark, T., Wolstenholme, J., Clarke, P.: Missing.... presumed at random: cost-analysis of incomplete data. Health Economics **12**(5), 377–392 (5 2003). https://doi.org/10.1002/hec.766, https://ideas.repec.org/a/wly/hlthec/v12y2003i5p377-392.html

5. Chen, J.F., Chen, W.L., Huang, C.P., Huang, S.H., Chen, A.P.: Financial time-series data analysis using deep convolutional neural networks. In: 2016 7th International Conference on Cloud Computing and Big Data (CCBD). pp. 87–92 (2016). https://doi.org/10.1109/CCBD.2016.027
6. Das, A., Kong, W., Sen, R., Zhou, Y.: A decoder-only foundation model for time-series forecasting (2024), https://arxiv.org/abs/2310.10688
7. Esling, P., Agon, C.: Time-series data mining. ACM Computing Surveys (CSUR) **45**, 12 (11 2012). https://doi.org/10.1145/2379776.2379788
8. Friedman, J.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics **29** (11 2000). https://doi.org/10.1214/aos/1013203451
9. Garza, A., Challu, C., Mergenthaler-Canseco, M.: Timegpt-1 (2024), https://arxiv.org/abs/2310.03589
10. Gaudiani, A.A.: Numerical methods for engineers (fifth edition). steven chapra and raymond canale. mc graw hill, 2005. Journal of computer science and technology (La Plata) **8**(2), 127–128 (2008)
11. Herzen, J., Lässig, F., Piazzetta, S.G., Neuer, T., Tafti, L., Raille, G., Pottelbergh, T.V., Pasieka, M., Skrodzki, A., Huguenin, N., Dumonal, M., Kościsz, J., Bader, D., Gusset, F., Benheddi, M., Williamson, C., Kosinski, M., Petrik, M., Grosch, G.: Darts: User-friendly modern machine learning for time series (2022), https://arxiv.org/abs/2110.03224
12. Hoyer, S., Hamman, J.: xarray: N-d labeled arrays and datasets in python. Journal of Open Research Software **5** (04 2017). https://doi.org/10.5334/jors.148
13. Jaiswal, J.K., Samikannu, R.: Application of random forest algorithm on feature subset selection and classification and regression. In: 2017 World Congress on Computing and Communication Technologies (WCCCT). pp. 65–68 (2017). https://doi.org/10.1109/WCCCT.2016.25
14. Liu, H., van Oosterom, P., Hu, C., Wang, W.: Managing large multidimensional array hydrologic datasets: A case study comparing netcdf and scidb. Procedia Engineering **154**, 207–214 (2016). https://doi.org/https://doi.org/10.1016/j.proeng.2016.07.449, https://www.sciencedirect.com/science/article/pii/S1877705816318380, 12th International Conference on Hydroinformatics (HIC 2016) - Smart Water for the Future
15. Meyer, M., Zapata, D., Kaltenpoth, S., Müller, O.: Benchmarking time series foundation models for short-term household electricity load forecasting (2024), https://arxiv.org/abs/2410.09487
16. Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J.: A time series is worth 64 words: Long-term forecasting with transformers (2023), https://arxiv.org/abs/2211.14730
17. Petneházi, G.: Recurrent neural networks for time series forecasting (12 2018). https://doi.org/10.48550/arXiv.1901.00069
18. Ye, J., Zhang, W., Yi, K., Yu, Y., Li, Z., Li, J., Tsung, F.: A survey of time series foundation models: Generalizing time series representation with large language model (2024), https://arxiv.org/abs/2405.02358
19. Young, T., Hazarika, D., Poria, S., Cambria, E.: Recent trends in deep learning based natural language processing [review article]. IEEE Computational Intelligence Magazine **13**(3), 55–75 (2018). https://doi.org/10.1109/MCI.2018.2840738
20. Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., Zhang, W.: Informer: Beyond efficient transformer for long sequence time-series forecasting (2021), https://arxiv.org/abs/2012.07436